

Work Order — Labor & Machine Costing

Kis product ko banane ki asli cost kya hai — material ke saath-saath labor aur machine cost bhi

CRM Module: Purchase & Production

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Status: Live / Built & Verified

1. Ye Feature Kya Hai

Work Order module pehle se hi raw material se finished good banane ka process track karta tha (kitna material use hua, finished good ka stock kitna badha). Lekin usme sirf **material ki cost** pata chalti thi — labor (mazdoori) aur machine (bijli, wear-and-tear, rent) ki cost kahin record nahi hoti thi.

Is addition ke baad, har Work Order ke "Production Cost" card me teen cheezein dikhti hain: **Material Cost** (automatic), **Labor Cost** aur **Machine Cost** (aap manually daalte ho), aur inka jod — **Total Cost** aur **Cost per Unit** (automatic calculate hota hai).

EK LINE ME

Ab aap ko pata chalega ki ek product banane me **asli kitna kharcha** aaya — sirf material nahi, poori production cost.

2. Kaun Use Karega, Kyun, Aur Kab

KAUN

Wo user jiske paas **work_orders.manage** permission hai — normally **Tenant Admin** ya production/procurement manager. Shop-floor staff jo sirf Work Order Start/Complete karta hai, use bhi ye permission chahiye hoga cost update karne ke liye.

KYUN

Sirf material cost dekh kar pata nahi chalta ki product banane me **asli profit margin** kya hai. Agar Steel Chair ka material cost ₹2,500 hai lekin usme ₹1,000 labor aur ₹500 machine bhi laga, to asli cost ₹4,000 hai — agar aap ₹3,500 me bech rahe ho to **loss** ho raha hai, sirf material dekh kar ye pata nahi chalta.

KAB

Work Order **complete** hone ke baad — jab actual labor hours aur machine usage ka pata chal jaye (timesheet, machine log, ya andaza se). Ye field kisi bhi status par edit ho sakta hai, isliye "abhi Work Order complete kiya, kal actual labor cost pata chala" — dono scenario me kaam karta hai.

DESIGN DECISION

Cost entry **jaan-boojh kar** Work Order ke status se independent rakha gaya hai (Pending/In Progress/Completed — kisi bhi status me edit ho sakta hai). Kyunki real life me actual

labor/machine cost aksar production complete hone ke *kai din baad* pata chalta hai (jab bill/timesheet aata hai), na ki completion ke exact waqt.

3. Kaise Kaam Karta Hai

- 1 Material Cost — automatic.** Product ke Bill of Materials (BOM) se calculate hota hai: har raw material ki *quantity-per-unit* × *Work Order ki quantity* × *material ka rate*, sabka jod. Isme kuch bhi type nahi karna — system khud nikal leta hai.
- 2 Labor Cost — manual entry.** Work Order ke show page par ek input box hai, jisme total labor cost (₹) daal sakte ho.
- 3 Machine Cost — manual entry.** Isi tarah, machine/electricity/overhead cost ke liye ek alag input box.
- 4 Total Cost aur Cost/Unit — automatic.** "Save Costs" dabate hi Total (Material + Labor + Machine) aur ek unit ki cost turant calculate ho jaati hai.

Example (jo actually test kiya gaya)

10 Steel Chair banaye, jinme har chair ke liye 5 Steel Rod (₹50/rod) chahiye:

MATERIAL COST	₹2,500.00
+ LABOR COST	₹800.00
+ MACHINE COST	₹200.00
TOTAL COST	₹3,500.00
÷ 10 UNITS = COST PER UNIT	₹350.00

4. Kahan Milega

Work Orders → koi bhi Work Order kholo → show page → "Production Cost" card (right-side sidebar me, Production actions ke bagal me). Yahi se cost dekhna aur edit karna dono ho sakta hai.

5. Abhi Ki Limitation

DHYAN RAKHEIN

Material Cost **"live" calculate** hota hai — matlab agar aap BOM ya material ka rate baad me change kar do, to purane Work Orders ka Material Cost bhi (jab dobara khologe) naye rate se dikhega, purane rate se nahi. Ye historical accuracy ke liye perfect nahi hai (asli accounting-grade costing me har Work Order apna "us waqt ka" rate snapshot rakhta hai), lekin abhi ke liye quick cost-estimate ke liye kaafi hai. Neeche "Future Enhancements" me isko fix karne ka option diya hai.

6. Aage Kya Add Kiya Ja Sakta Hai

Ye poori list hai — sab ek saath karne ki zaroorat nahi. Har cheez ki apni value aur effort hai:

FEATURE	KYA KAREGA	VALUE	EFFORT
Historical Cost Snapshot	Material Cost ko Work Order complete hote hi "freeze" kar do (jaise Quotation/Invoice apne items snapshot karte hain) — taaki baad me rate change hone se purane records na badlein.	High	Small
Per-Hour Labor Rate	Flat "Labor Cost" ki jagah "Hours Worked × Rate/Hour" — zyada granular aur reusable across Work Orders (ek baar rate set karo, sirf hours badalte rahein).	Medium	Small
Margin Analysis	Product ki selling rate (Invoice/Quotation wala) minus production Cost/Unit = asli profit margin, dashboard par dikhna.	High	Medium

Cost Variance Report	Estimate vs Actual cost compare karna across multiple Work Orders — pata chale konsa product consistently expected se mehenga ban raha hai.	Medium	Medium
Multi-level BOM	Ek raw material khud kisi aur raw material se bane (sub-assembly) — tabhi zaroori hai agar aapka production multi-step hai.	Situational	Large
Shop-Floor Tracking	Kaunsa operator, kaunsi machine/station, kitna time laga — sirf tab zaroori jab literally multiple workstations manage karni ho.	Situational	Large
Overhead Allocation %	Machine cost manually daalne ki bajaye, ek fixed % (material cost ka) automatically overhead ke roop me add ho jaye.	Low	Small

Ye document is CRM ke Work Order Costing feature (built & verified 20 August 2026) ka reference hai. Feature abhi live hai aur poore automated test suite (180/180) ke saath verified hai.